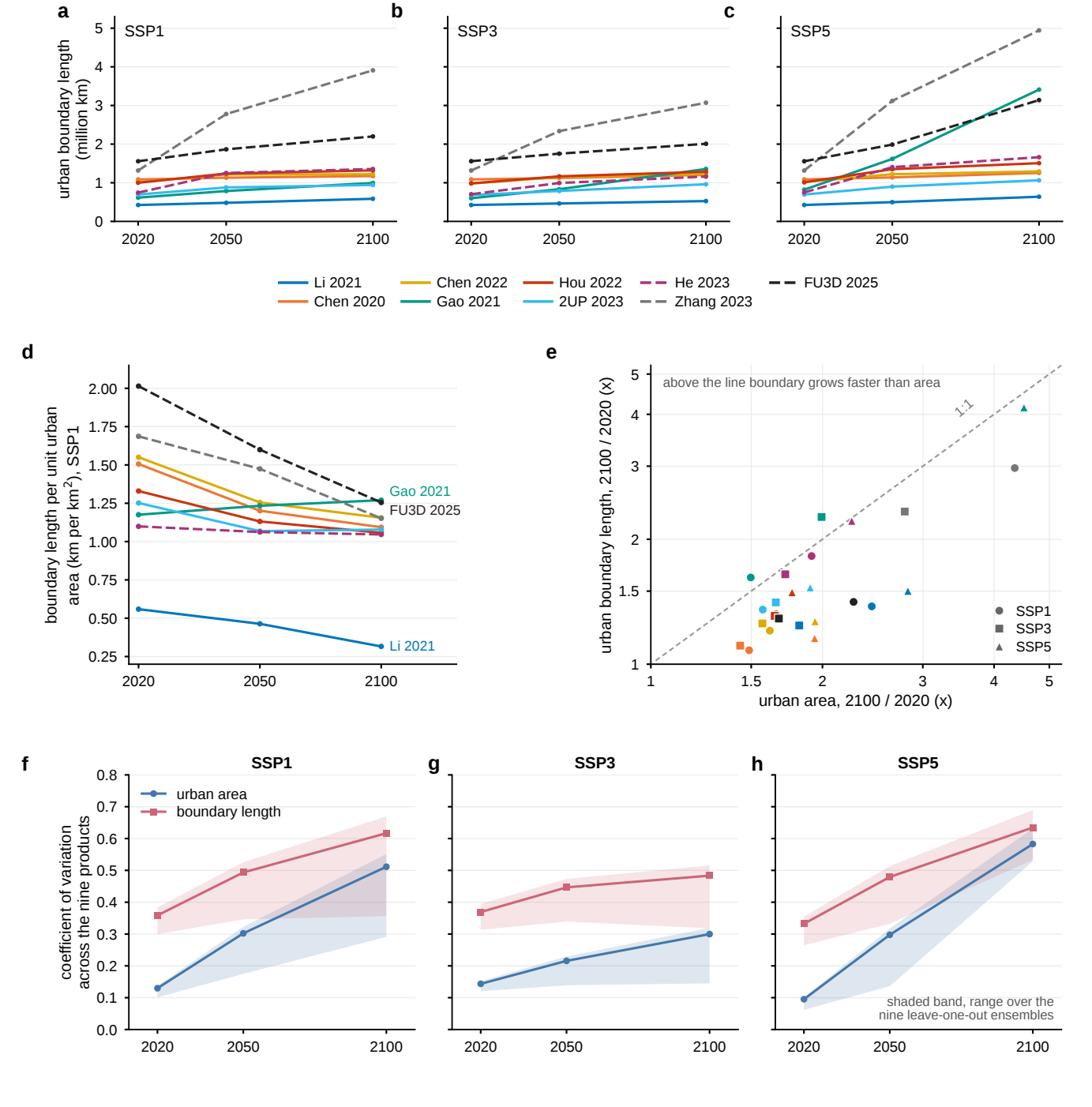
a
b
c
SSP1
SSP3
SSP5
urban boundary length (million km)
0
1
2
3
4
5
2020
2050
2100
Li 2021
Chen 2020
Chen 2022
Gao 2021
Hou 2022
2UP 2023
He 2023
Zhang 2023
FU3D 2025
d
boundary length per unit urban area (km per km^2), SSP1
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Gao 2021
FU3D 2025
Li 2021
e
above the line boundary grows faster than area
1:1
urban boundary length, 2100 / 2020 (x)
urban area, 2100 / 2020 (x)
1
1.5
2
3
4
5
SSP1
SSP3
SSP5
f
g
h
SSP1
SSP3
SSP5
urban area
boundary length
coefficient of variation across the nine products
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
shaded band, range over the nine leave-one-out ensembles